Project Proposal: Graph-Based Risk Mapping: Mining Movement Networks to Predict Vehicular Mortality in Florida Panthers

George Alenchery
B.S. Software Engineering
U.A. Whitaker College of Engineering, FGCU

Jeremy Martin
B.S. Software Engineering
U.A. Whitaker College of Engineering, FGCU

Marcelo Amorim
B.S. Software Engineering
U.A. Whitaker College of Engineering, FGCU

Department of Computing and Software Engineering
Florida Gulf Coast University
Fort Myers, FL, USA

Research Question

Can a graph neural-network trained on telemetry-derived movement connectivity outperform non-spatial baseline models in predicting road segments at high risk of Florida panther vehicular mortality?

Gap in Existing Research

Wildlife-vehicle collisions (WVC) numbers are still high even though we have great advancements in the technology of the automotive industry (Pagany, 2022). A study conducted in Bavaria was conducted with a dataset of around 731k accidents reported, implementing Random Forest (RF), FeedForward Neural Networks (FNN) and Support Vector Machine, with samples of accidents reported varying from 2010 to 2019 (Pagany, 2022). Non-accidents were randomly generated in python over time and space for helping with the supervised, and the final results showed Random Forests with 86.6% RPQ and FeedForward neural networks with 86.7% RPQ (risk prediction quality).

Crossing structures are one commonly used method for reducing wildlife–vehicle collisions but limiting funding by governments and stakeholders reduces the amount of structures that can be constructed (Loraamm & Downs, 2015). A research applied probabilistic time-geographic strategy coupled with site selection phases handled by classical optimization model to site wildlife crossing structures. Probabilistic Space–Time Prism (STP) calculations were implemented with a custom Python script (Loraamm & Downs, 2015). The goal is to cover the largest area having the highest probability of interaction given a finite number of crossing structures that are available to be located. Coverage means fencing distance associated with that particular structure, and the approach was demonstrated using Florida panther telemetry data

identifying potential crossing structures across two counties in south Florida (Loraamm & Downs, 2015). Maximal covering location problem was solved for four coverage distances while utilizing radio telemetry as tracking data, capturing frequent contact with roads. Results point to the most effective coverage distance being 2km, which incrementally covers more total animal–road interaction probability than that of lower fencing distances in the case of the Florida panther.

Traffic accident analysis on road networks based on road network connections and traffic volume was covered for a study across the US (Nippani et al, 2023). Currently, there is a lack of publicly available accident datasets for comprehensive evaluations. A large-scale, unified dataset of traffic accident records from official reports of various states in the US (9 million in total) was constructed, accompanied by road networks and traffic volume reports (Nippani et al, 2023). Using this construction existing deep-learning methods were evaluated for predicting the occurrence of accidents on road networks. GraphSAGE showed to be accurate in predicting the number of accidents on roads, averaged over states. Multitask learning was used for accounting for cross-state variabilities and transfer learning to combine traffic volume with accident prediction (Nippani et al, 2023).

Existing studies have separately applied machine learning to wildlife-vehicle collision prediction, graph neural networks to general road-accident prediction, and telemetry-based optimization to Florida panther crossing analysis. However, among the studies reviewed, these approaches have not been combined to incorporate telemetry-derived movement connectivity into a graph neural network for predicting road-segment-level Florida panther vehicular mortality. Based on the previously cited works, this shows as a promising approach to the field.

Datasets

Florida Panther Mortality
(Florida_Panther_Mortality.csv & Florida_Panther_Mortality.geoJSON).
The FWC Panther Pulse mortality database (Florida Fish and Wildlife Conservation Commission 2025a), publicly available with 691 records spanning 1949 to 2025. Each record contains precise coordinates in Florida State Plane, date, sex, age, cause of death, county, and a free-text location description. We restrict our research to vehicular deaths from 2000 onward, yielding approximately 350 records.

Florida Panther Telemetry
(Florida_Panther_Telemetry.csv & Florida_Panther_Telemetry.geoJSON).
The FWC Florida Panther Telemetry dataset (Florida Fish and Wildlife Conservation Commission 2025b), publicly available with 112,802 records spanning February 1981 to January

2025 across 276 individual panthers. Collected by FWC, Big Cypress National Preserve, and Everglades National Park using aerial VHF radio tracking following the methodology of Land et al. (2008). Most fixes are diurnal resting locations. Panther identifiers (CATNUMBER) are compatible with mortality records (PANTHERID), enabling per-individual analysis for panthers appearing in both datasets. We again restrict our research to datapoints dated 2000 onward, yielding approximately 63,170 records.

Road network graph.
(LRS Routes with Measures TDA)
The road network is derived from the FDOT LRS Routes with Measures layer (Transportation Data & Analytics), a geometric representation of all roadways in the Roadway Characteristics Inventory (RCI), including both the State Highway System and off-system county roads, with segment-level attributes for posted speed limit, functional classification, and annual average daily traffic. Publicly available from the FDOT GIS portal and Open Data Hub.

Data Selection/Scoping

Datasets

The Florida Fish and Wildlife Conservation Commission (FWC) attributes the rise in panther roadkill to growth in the panther population itself, reporting "As panther numbers increased over time, the number of panther roadkills has also been on the rise. Since 2000, the annual number of roadkills has ranged from six to 34." [1]. Figure 1, taken from FWC's wildlife crossings page, shows this trend.

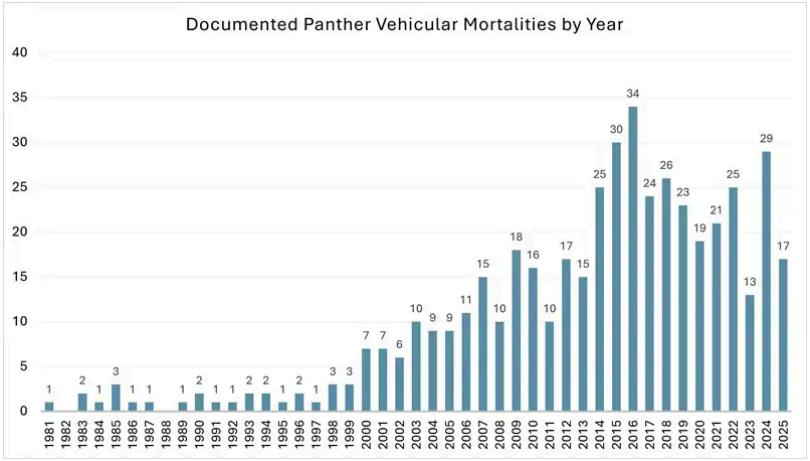


Figure 1: Number of Panther Vehicular Mortalities (1981-2025)

The mortality dataset used in this study, shown in *Figure 2,* reproduces that range exactly, recording a range of six vehicular deaths to a maximum of 34 between the years 2000-2016. The yearly distribution is strongly right-skewed, with only 11% of the data points recorded for

1981–1999, against the 89% for 2000–2025 as seen in *Figure 3*. This shift reflects a genuine increase in panther-vehicle conflict as the population expanded into a more developed landscape, and it motivates restricting analysis to the post-2000 period where records are densest.

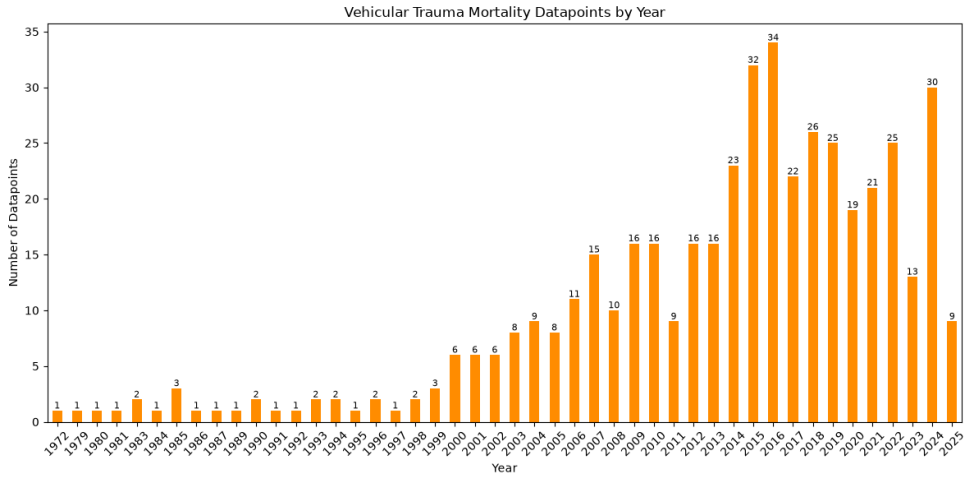


Figure 2: Mortality Datapoints by Year (Including ALL mortalities)

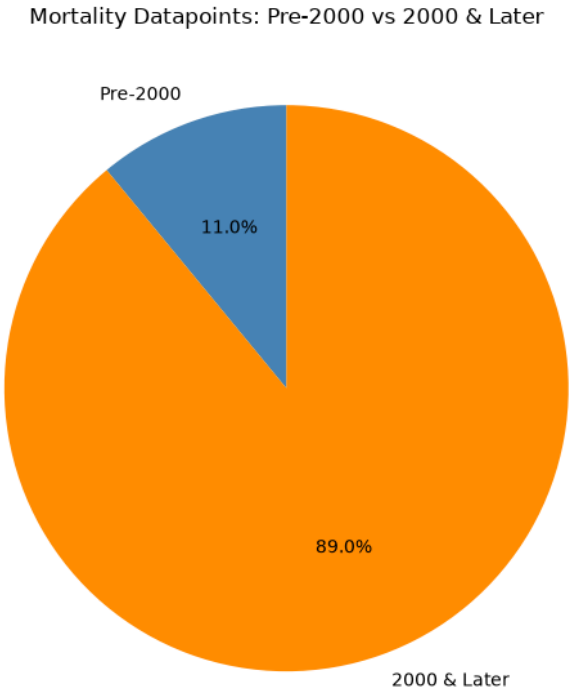


Figure 3: Mortality Datapoints Pre-2000 vs. Post-2000 & Later

Study area.

Collier, *Hendry*, *Lee* and *Glades* counties, which together account for 92.3% of recorded vehicle mortality events, as seen in *Figure 4*. For simplicity we will scope our records to these counties.

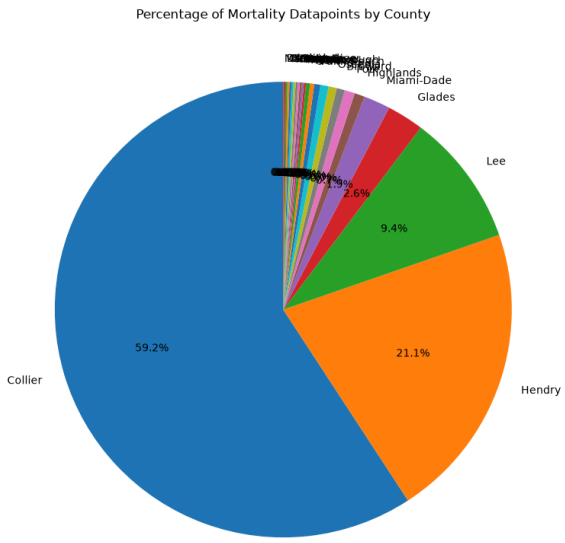


Figure 4: Percentage of Mortality Datapoints by County

Proposed Methodology

The three datasets will be reprojected to a common CRS and filtered to observations from 2000 onward. Rows with missing or invalid coordinates will be dropped and date fields standardized across datasets. Telemetry uses UTM Zone 17N (NAD83) natively; mortality includes both UTM and latitude/longitude. The identifier formats between telemetry (CATNUMBER) and mortality (PANTHERID) will be inspected for compatibility, though individual level linkage is not required for the primary analysis.

The study area will be represented as a graph where each node is a road segment split at intersections. Road segments were chosen over grid cells because a segment maps directly to what a wildlife agency or transportation planner would act on and because FDOT attributes like speed limit and traffic volume are already defined at that resolution.

Bringing telemetry into the graph, each panther's fixes will be sorted by date and consecutive fixes connected by straight lines. This, however, is an approximation and not necessarily the exact path that the panther may have walked. More sophisticated movement models exist but add scope beyond this project's timeline and resources.

Wherever an interpolated segment crosses a road, a candidate crossing point is recorded and attributed to that road segment. Aggregating across all panthers gives a crossing frequency per road segment, which serves as both a node feature and a signal for adding movement-derived edges between segments that share panther traffic. The final graph combines these movement-derived edges with standard spatial adjacency between physically-connected road segments.

Node features will include telemetry-derived crossing frequency and telemetry point density alongside FDOT attributes like speed limit, AADT, road classification and length. A segment is labeled positive if a recorded vehicle mortality falls within a buffer distance of it; buffer values between 100 and 500 meters will be tested, with the final choice sanity checked against the free-text location descriptions in the mortality dataset where they include mile markers or road names.

Three model families will be compared. First, non-spatial baselines (random forest and XGBoost) trained on the same feature table with no graph structure. Second, a kernel density baseline that projects historical strikes directly onto road segments. Third, the graph model, using GraphSAGE for its inductive properties, which matter for eventually scoring newly built or proposed roads. All models will use identical features and identical cross-validation splits so that any performance difference can be attributed to graph structure rather than preprocessing choices. Class imbalance will be handled with weighted loss functions and evaluation metrics designed for imbalanced data (AUC-PR and Brier score rather than raw accuracy).

Evaluation Plan

Success will be measured against the central research claim that the graph model outperforms non-graph baselines that use the same features without graph structure. If this holds true, the performance difference is attributable to graph structure rather than preprocessing or feature choices.

Primary metrics are AUC-PR (area under the precision-recall curve) and precision-at-K for a range of K values reflecting practical intervention scales. Brier score will be reported for calibration. Raw accuracy will not be reported, since a trivial model predicting no strikes everywhere would score above 95% on this imbalanced data.

Cross-validation will use a spatial scheme to prevent leakage between adjacent road segments that share graph structure. The specific spatial scheme will be selected during implementation based on empirical fold characteristics and the trade-off between fold count and spatial separation. Random k-fold splits are not appropriate for graph-based spatial models. All models will be trained and evaluated on identical folds, with performance reported as mean and standard

deviation across folds. Paired comparisons will be used to assess whether the graph model's advantage over baselines is consistent rather than driven by a single fold.

References

[1] *Florida Fish and Wildlife Conservation Commission, "Wildlife Crossings." [Online]. Available: https://myfwc.com/wildlifehabitats/wildlife/panther/wildlife-crossings/. [Accessed: Sep. 5, 2026].*

R. W. Loraamm and J. A. Downs, "A wildlife movement approach to optimally locate wildlife crossing structures," *International Journal of Geographical Information Science*, vol. 30, no. 1, pp. 74–88, 2016, doi: 10.1080/13658816.2015.1083995.

A. Nippani, D. Li, H. Ju, H. N. Koutsopoulos, and H. R. Zhang, "Graph neural networks for road safety modeling: Datasets and evaluations for accident analysis," *arXiv:2311.00164*, 2023, doi: 10.48550/arXiv.2311.00164.

R. Pagany, "A spatiotemporal risk prediction of wildlife-vehicle collisions using machine learning for dynamic warnings," *Journal of Safety Research*, vol. 83, pp. 269–281, Dec. 2022, doi: 10.1016/j.jsr.2022.09.001.